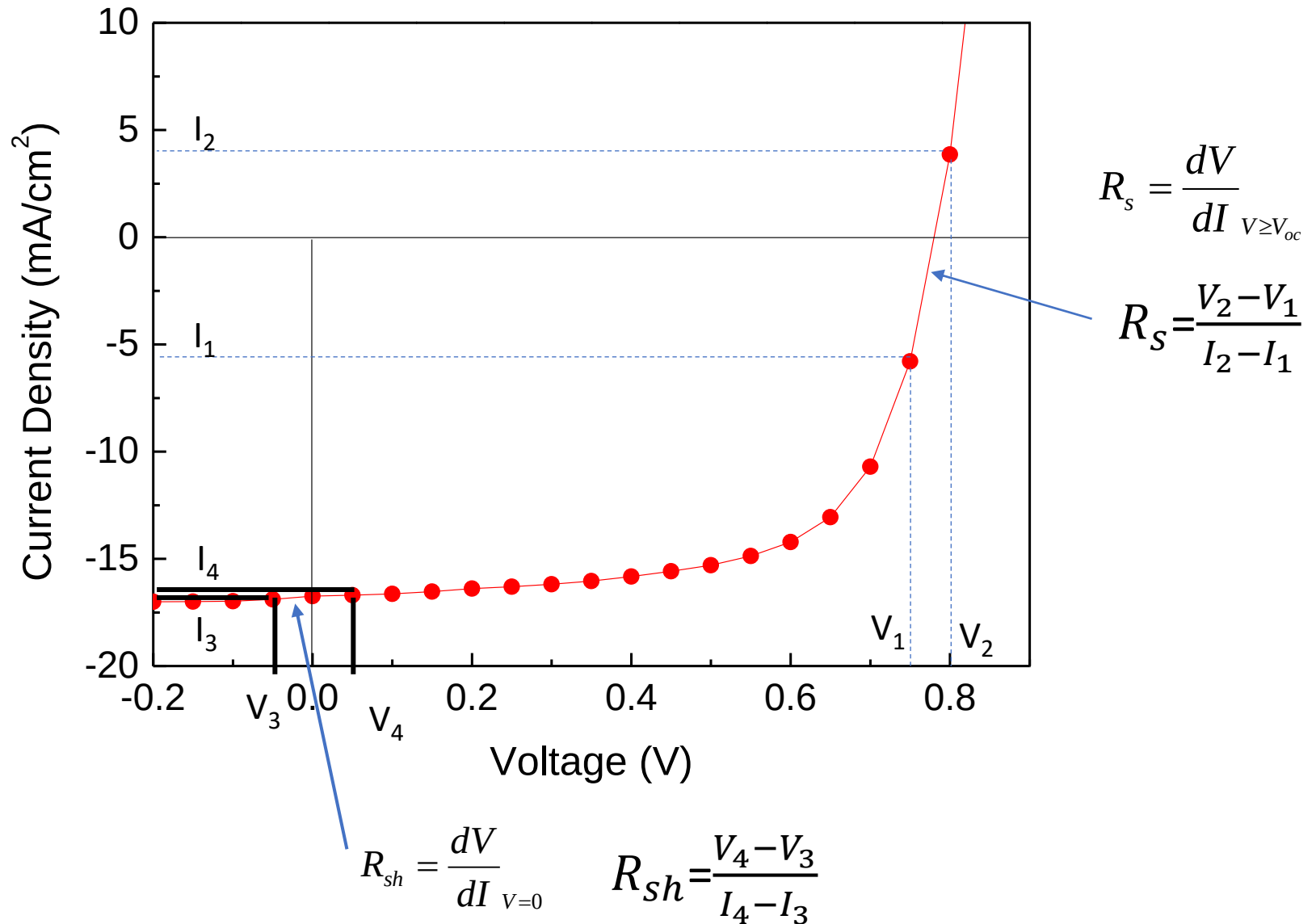


# Calculation of Rs and Rsh from J-V curve



## Extraction of n and $I_0$ from dark J-V curve

$$I = I_0 \exp\left(\frac{qV}{nKT}\right)$$

$$\ln I = \frac{qV}{nkT} + \ln I_0$$

According to the slope equation (  $y = mx + c$  ),

$$\text{Slope} = \frac{\ln I}{V} = \frac{1}{\frac{nkT}{q}}, \text{ y-axis intersection point is } \ln I_0.$$

$$n = \frac{1}{\text{slope} \times kT/q} \text{ (} kT/q = 0.026 \text{ eV at 300 K)}$$

